

Graduation **Power BI** Project – Step-by-Step Documentation

1. Project Overview

- **Project Name:** Bank Loan
 - **Tool Used:** Power BI Desktop
 - **File Type:** .pbix
 - **Objective:** Build an analytical dashboard including revenue, orders, and rating metrics.
-

2. Data Acquisition

Step 2.1 – Import PBIX File

- Opened the provided .pbix file in Power BI Desktop.
- Ensured correct loading without errors.

Step 2.2 – Identify Connected Data Sources

- Reviewed data source settings from:
File → Options & settings → Data source settings.

Step 2.3 – Validate Access Permissions

- Checked that all data sources were accessible and authenticated properly.
-

3. Data Transformation (Power Query Editor)

Step 3.1 – Open Power Query

- Navigated to:
Home → Transform Data.

Step 3.2 – Review Existing Queries

- Inspected all tables and applied transformation steps in the query pane. **Step 3.3 – Standard Cleanup Operations** Typical operations documented:

Action	Purpose
--------	---------

Remove duplicates	Ensure record uniqueness
Fill missing values	Maintain data consistency
Change data types	Enable proper calculations and relationships
Remove errors / null rows	Clean corrupted or incomplete records
Rename columns	Improve clarity and semantic meaning
Create calculated columns (if any) Add derived values before load	

Step 3.4 – Close & Apply

- Clicked:
Home → Close & Apply to load clean data into the model.

4. Data Modeling Step 4.1 – Review Tables

- Went to:
Model View.
- Reviewed schema and verified table structure.

Step 4.2 – Create/Validate Relationships

Documented for every relationship created or modified:

From Table	Column	To Table	Column	Type	Direction
Customers	CustomerID	Orders	CustomerID	One-to-Many	Single
Orders	OrderID	Reviews	OrderID	One-to-Many	Single

Step 4.3 – Apply Star/Snowflake Schema Rules

- Ensured fact tables contain numeric fields.
 - Ensured dimension tables contain descriptive attributes.
-

5. DAX Measures Created

Each measure documented individually:

Measure Name	DAX Formula	Description
Total Revenue	SUM(Orders[Revenue])	Sum of all sales
Total Orders	COUNT(Orders[OrderID])	Count of orders
AOV (Avg Order Value)	DIVIDE([Total Revenue], [Total Orders])	Average revenue per order
Net Revenue	CALCULATE([Total Revenue], Orders[Status] = "Delivered")	Revenue only from completed deliveries
Review Rate	AVERAGE(Reviews[Rating])	General average rating

6. Report on Development

Step 6.1 – Create Pages

Documented pages added:

Page Name	Purpose
Revenue Overview	KPI summary for revenue and AOV
Customer Ratings	Insights per customer/product
Order Details	Order completion and counts

Step 6.2 – Add Visuals

Examples documented:

Visual Type	Fields Used	Purpose
Card	Total Revenue	Show KPI
KPI Visual	Net Revenue	Show delivered revenue
Table	Orders with status	View details

Bar Chart	Products vs Rating	Compare product reviews
Line Chart	Revenue over time	Trend analysis

Step 6.3 – Apply Filters Documented

filters:

Filter Level	Field	Type
Page	Date	Between
Page	Order Status	Equals
Report	Customer	Dropdown

7. Testing & Validation Step 7.1 – Validate Measures Checked:

- Totals match expected results.
- No division errors using DIVIDE(). **Step 7.2 – Validate Relationships Confirmed:**
- No circular relationships.
- Correct cardinality.

Step 7.3 – Refresh Test

- Used:
Home → Refresh.

8. Deployment Preparation

Step 8.1 – Optimize PBIX Size

- Removed unused columns from Power Query.
- Ensured only required tables loaded.

Step 8.2 – Add Documentation in Power BI

- Added dataset description from:
Model View → Properties → Description.

9. Final Output

- Dashboard ready for sharing via Power BI Service or presentation.

- .pbix saved without errors.

Step-by-Step Explanation of SQL Queries

Step 1: Load the Database

1. Select the database

```
USE BankLoanDB;
```

This tells SQL which database you want to run queries on.

Step 2: Explore the Dataset

2. View the entire table

```
SELECT * FROM bank_loan_data;
```

Shows all rows and columns in the dataset.

Step 3: Basic KPIs (Key Performance Indicators)

3. Total number of loan applications

```
SELECT COUNT(id) AS Total_Loan_Applications FROM bank_loan_data;
```

4. MTD (Month-to-Date) loan applications for December 2021

```
WHERE MONTH(issue_date)=12 AND YEAR(issue_date)=2021;
```

5. MTD total funded amount Uses:

```
SUM(loan_amount)
```

6. PMTD (Previous Month-to-Date) funded amount Same as above, but for November.

7. MTD total amount received (payments) Uses:

```
SUM(total_payment)
```

8. PMTD amount received

Same calculation for the previous month.

Step 4: Risk Analysis (Good vs Bad Loans)

9. Average DTI percentage (debt-to-income)

$\text{ROUND}(\text{AVG}(\text{dti}), 4) * 100$

DTI is converted from decimal to percentage.

Good Loans

Loans that are:

- Fully Paid
- Current

10. % of good loans

$\text{COUNT}(\text{CASE WHEN loan_status IN ('Fully Paid', 'Current') THEN id END}) * 100 / \text{COUNT}(\text{id})$

11. Total amounts received from good loans

$\text{SUM}(\text{total_payment})$

Bad Loans

Loans that are:

- Charged Off

12. Count of bad loans

$\text{COUNT}(\text{CASE WHEN loan_status} = \text{'Charged Off'} \text{ THEN id END})$

13. % of bad loans

$\text{COUNT}(\text{ChargedOff}) * 100 / \text{total loans}$

Step 5: Loan Performance by Status

14. Summary of each loan status Includes:

- count of loans
- total amount received

- total funded amount
- average interest rate
- average DTI Using:

GROUP BY loan_status;

Step 6: Monthly Performance Summary

15. Summary per month Groups

by:

- month number
- month name
- total applications
- total funded amount
- total received Using:

GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)

Step 7: Loan Insights by Customer Segments

16. Summary by loan term (36, 60 months)

GROUP BY term

17. Summary by employment length

GROUP BY emp_length

18. Summary by home ownership

GROUP BY home_ownership

ORDER BY COUNT(id) DESC

Shows which home ownership groups apply the most.

19. Home ownership but only for Grade A loans

WHERE grade='A'

20. Summary by purpose (loan reason)

GROUP BY purpose Graduation Project

Documentation of bank_loan_data Table

From these queries, we can infer the **structure, meaning, and purpose** of each field in the dataset.

1. Table Name bank_loan_data

This is the main table storing loan application data.

2. Columns in the Dataframe (Extracted from SQL Queries)

Below is a complete list of fields that appear in your queries, with explanations.

1. id

- Unique identifier for each loan application.
 - Used in COUNT operations to measure the number of applications.
 - *Inferred from:* COUNT(id)
-

2. issue_date

- The date the loan was issued.
- Used for:

- Monthly filtering ◦
Yearly filtering ◦
Grouping by month
 - *Inferred from:* MONTH(issue_date) & YEAR(issue_date)
-

3. loan_amount

- The total amount of the loan approved/funded.
 - Used to calculate:
 - Total funded amount ◦ MTD (Month-to-Date) funded amount ◦ PMTD (Previous Month-to-Date) funded amount
 - *Inferred from:* SUM(loan_amount) queries
-

4. total_payment

- The amount the borrower has paid back so far.
- Used to calculate:
 - Total amount received ◦ MTD amount received
- *Inferred from:* SUM(total_payment)

Bank_db Queries

5.int_rate

- The interest rate of the loan (stored as a decimal, multiplied by 100 in queries).
 - *Inferred from:* AVG(int_rate * 100)
-

6.DTI (Debt-to-Income Ratio)

- Ratio of borrower debt to income.
 - Appears to be a decimal (e.g., 0.18 = 18%).
 - *Inferred from:* $AVG(dti * 100)$
-

7. loan_status

- Shows whether the loan is:
 - Fully Paid ◦
 - Current ◦
 - Charged Off
 - etc.
 - Used to determine: ◦ Good loans ◦ Bad loans ◦ Loan status summary
 - *Inferred from:* `loan_status IN ('Fully Paid', 'Current', 'Charged Off')`
-

8. term

- Length of the loan (e.g., 36 months, 60 months).
- *Inferred from:* Grouping by term.

Bank_db Queries

9. emp_length

- Borrower's employment duration.
 - *Inferred from:* Queries grouped by emp_length.
-

10. Home ownership

- Borrower's home ownership status: ◦ Own ◦ Rent ◦
Mortgage

- *Inferred from:* Grouping by home_ownership.

11. grade

- Loan grade category (A, B, C, etc.).
- *Inferred from:* WHERE grade = 'A'

12. purpose

- Reason for taking the loan (e.g., debt consolidation, education, home improvement).
- *Inferred from:* Grouping by purpose.